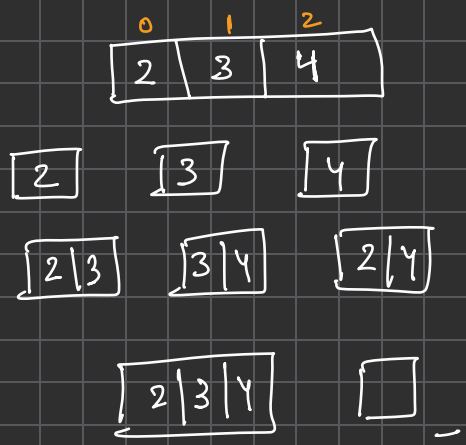



$2^3 = 8$

2
3
4
5
7
6
9
0

return

$2^4 \Rightarrow \boxed{16}$

$\boxed{1 \mid 4 \mid 3 \mid 9}$

$\boxed{1}$ $\boxed{4}$ $\boxed{3}$ $\boxed{9}$

$\boxed{1 \mid 4}$ $\boxed{4 \mid 3}$ $\boxed{3 \mid 9}$ $\boxed{1 \mid 3}$ $\boxed{1 \mid 9}$

$\boxed{4 \mid 9}$

$\boxed{1 \mid 4 \mid 3}$ $\boxed{1 \mid 4 \mid 9}$ $\boxed{1 \mid 3 \mid 9}$

$\boxed{4 \mid 3 \mid 9}$

$\boxed{1 \mid 4 \mid 3 \mid 9}$ $\boxed{\phantom{1}}$

| 0 | 1 | 2 |
|---|---|---|
| 2 | 3 | 4 |

$\boxed{\phantom{0}}$

$\boxed{2}$

$\boxed{3}$

$\boxed{4}$

$\boxed{2 \mid 3}$

$\boxed{2 \mid 4}$

$\boxed{3 \mid 4}$

$\boxed{2 \mid 3 \mid 4}$

| 2 | 3 | 4 |
|---|---|---|
| ✓ | ✗ | ✗ |

| 2 | 3 | 4 |
|---|---|---|
| ✗ | ✗ | ✗ |

| 2 | 3 | 4 |
|---|---|---|
| ✗ | ✓ | ✗ |

| 2 | 3 | 4 |
|---|---|---|
| ✗ | ✓ | ✓ |

| 2 | 3 | 4 | |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 4 |
| 0 | 1 | 0 | 3 |
| 0 | 1 | 1 | 7 |
| 1 | 0 | 0 | 2 |
| 1 | 0 | 1 | 6 |
| 1 | 1 | 0 | 5 |
| 1 | 1 | 1 | 9 |

$3 \Rightarrow 8$ ✓

$2^3$

$2^n$ ✓

$4 = 2^4 \Rightarrow 16$

$5 = 2^5 \Rightarrow ?$

2 3 4

2 3

2 4

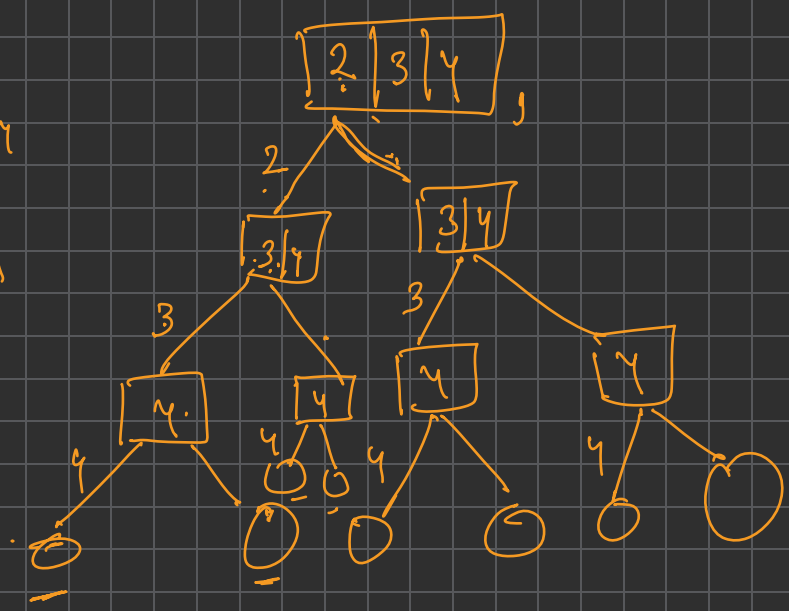
2

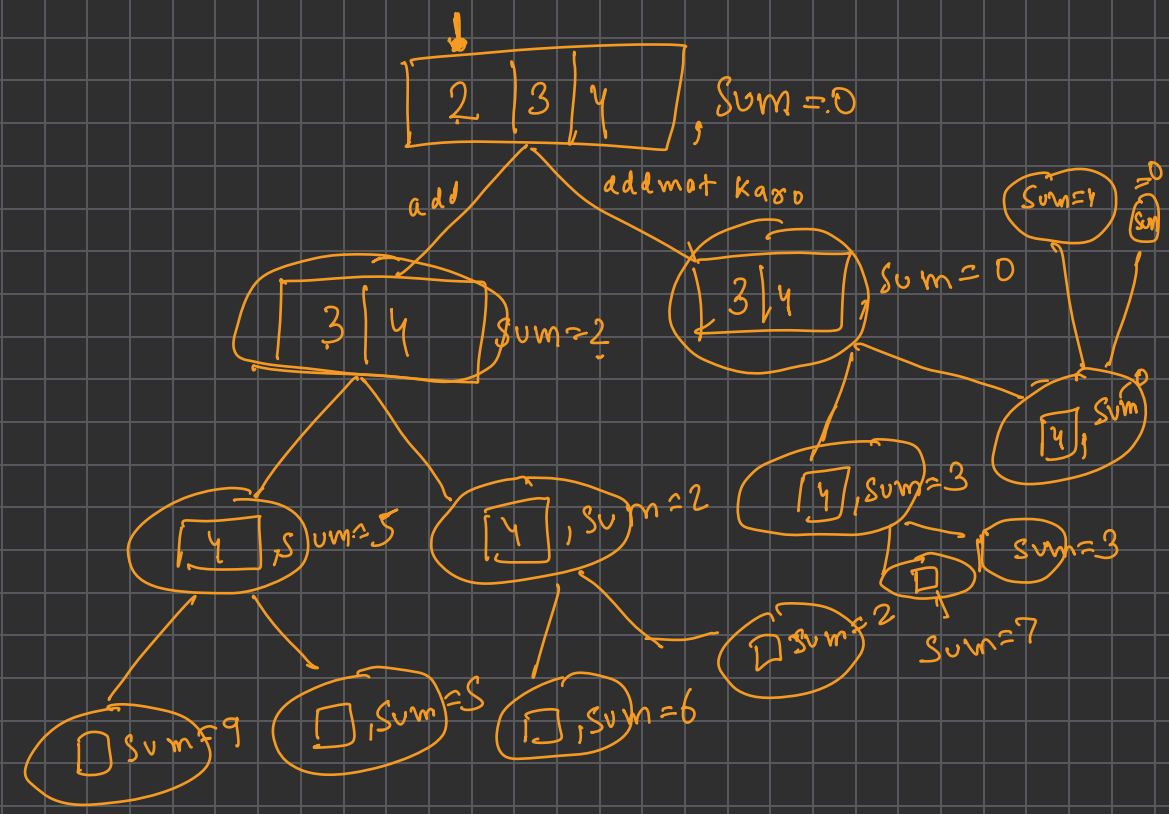
3 4

3

4

0




2 | 3 | 4 , Sum = 0
add
addmat karo
3 | 4 Sum = 2
3 | 4 Sum = 0
Sum = 4
=0
Sum
4 , Sum
4 , Sum = 5
4 , Sum = 2
4 , Sum = 3
Sum = 3
Sum = 2
Sum = 7
Sum = 9
, Sum = 5
, Sum = 6

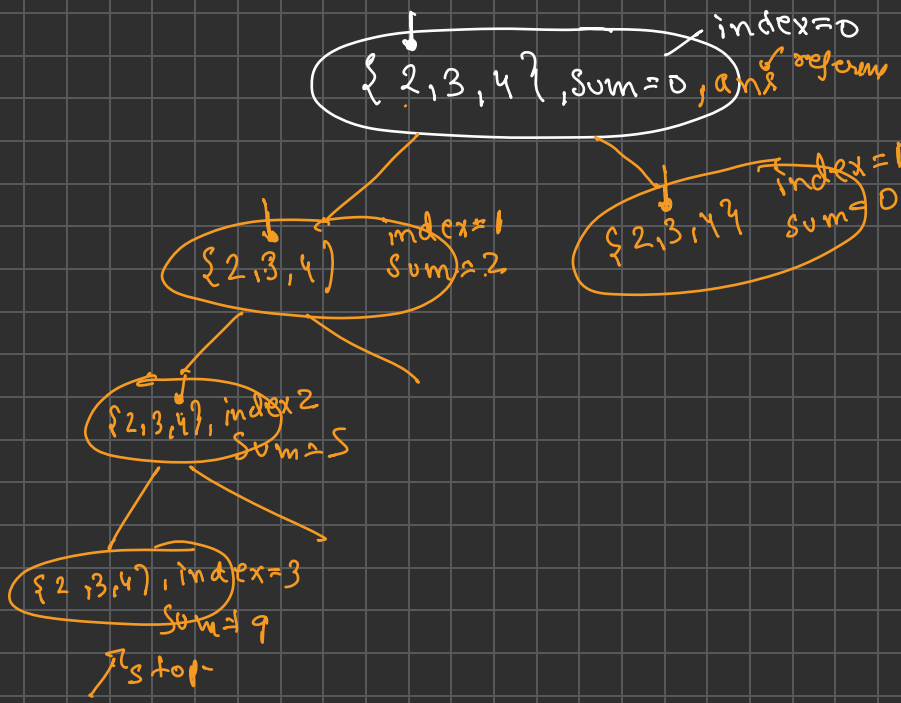
| | | |
|---|---|---|
| 9 | 2 | 4 |
| 5 | 7 | 0 |
| 6 | 3 | |

| 0 | 1 | 2 |
|---|---|---|
| 2 | 3 | 4 |

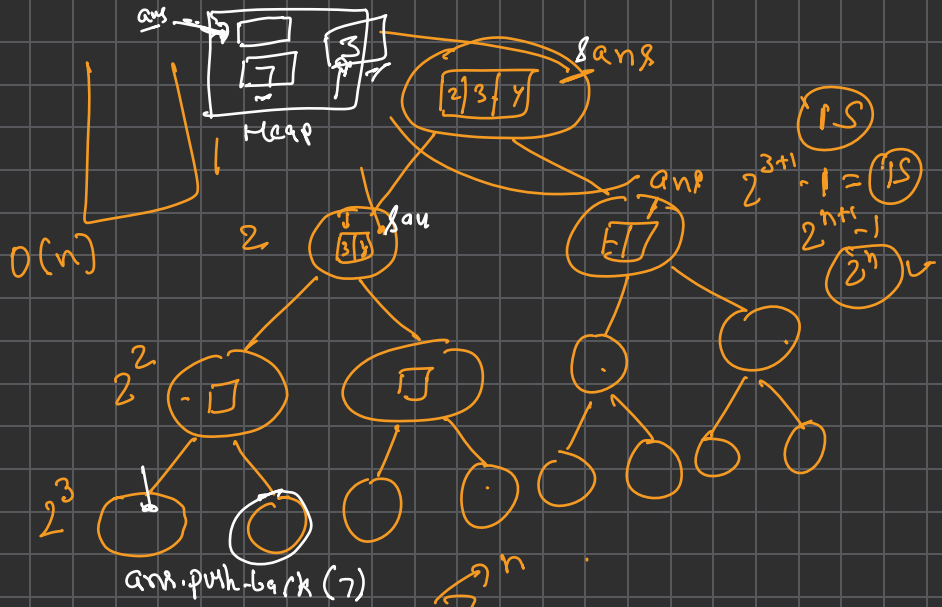
index = 0
sum = add
sum = -

```
void    subsetSum( vector<int>& arr , int index, int n , int sum
                   vector<int>& ans )
{
        if (index == n)
        {   ans.push_back(sum);
            return ;
        }

        subsetSum(arr, index+1, n, sum+ arr[index], ans)
        subsetSum( arr ,index+1, n, sum , ans )
}
```


{2,3,4}, sum=0, ans
index=0
return
{2,3,4}
index=1
sum=2
{2,3,4}
index=1
sum=0
{2,3,4}, index 2
sum=5
{2,3,4}, index=3
sum=9
stop

8 9




$$\left[2^0+2^1+2^2+2^3\right] \to n$$

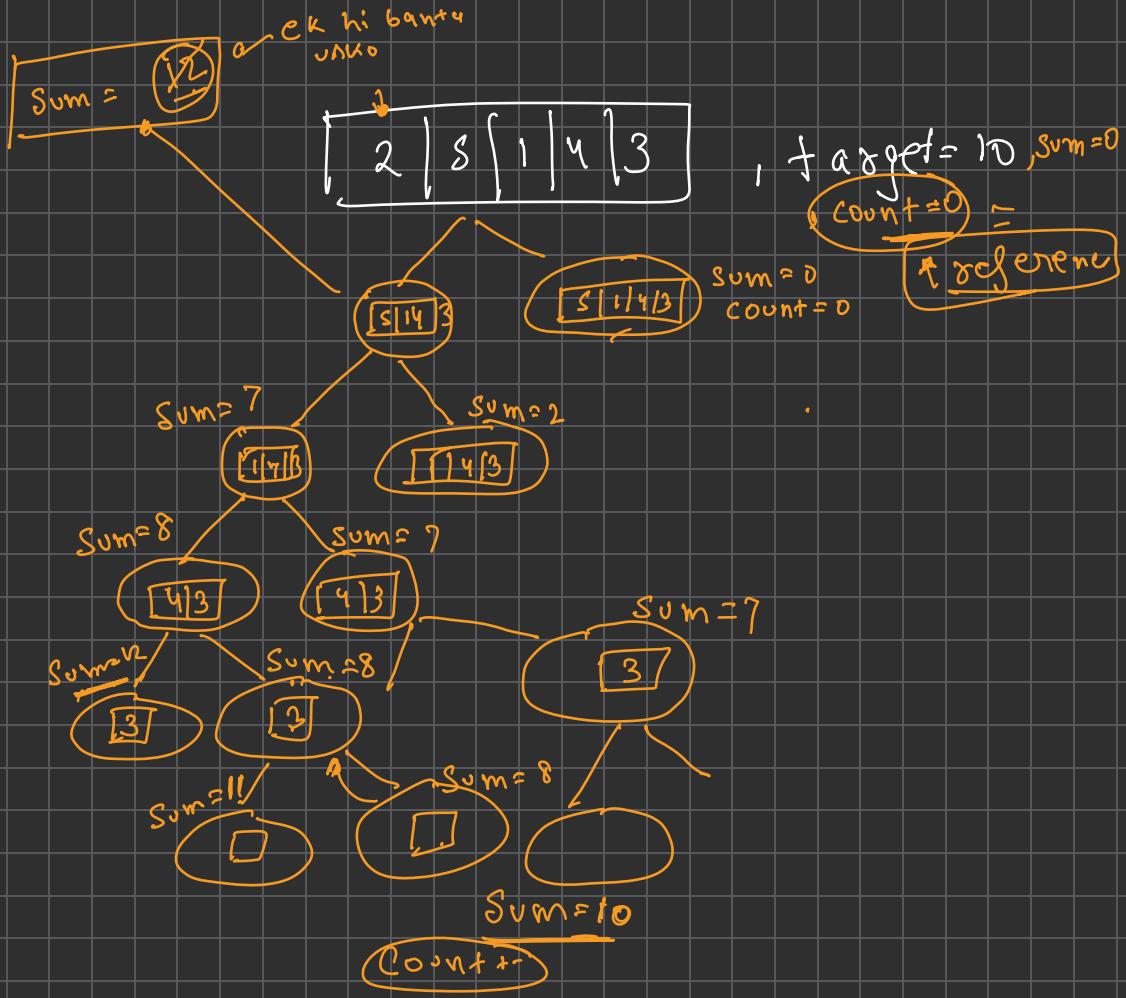
$$\left[2^0+2^1+2^2+\cdots 2^n\right]$$

$$2^{n+1}-1 \Rightarrow 2^n$$

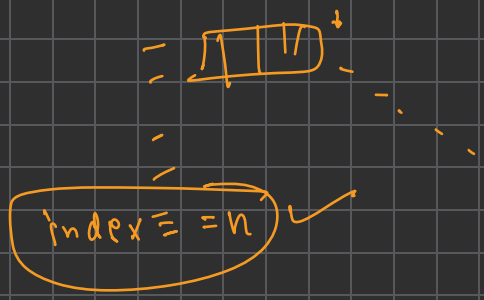
vector<int>ans

$$\left[\text{Space Complexity} = O(n) + 2^n\right]$$

$$= 2^n \rightarrow \boxed{\text{Space complexity}}$$




```
void FindSubset(vector<int>& arr, int index, int n,
                int sum, int &count) target)
{
    //
    if (sum==target)
    {
        count++;
        return;
    }
    if (sum > target)
        return;
    FindSubset(arr, index+1, n, sum+arr[index], count, tar)
    FindSubset(arr, index+1, n, sum, count, target)
}
```


index == n

{2, 5, 3}

| 2 | 5 | 1 | 4 | 3 |
|---|---|---|---|---|

index=1
sum=2

index=2
sum=7

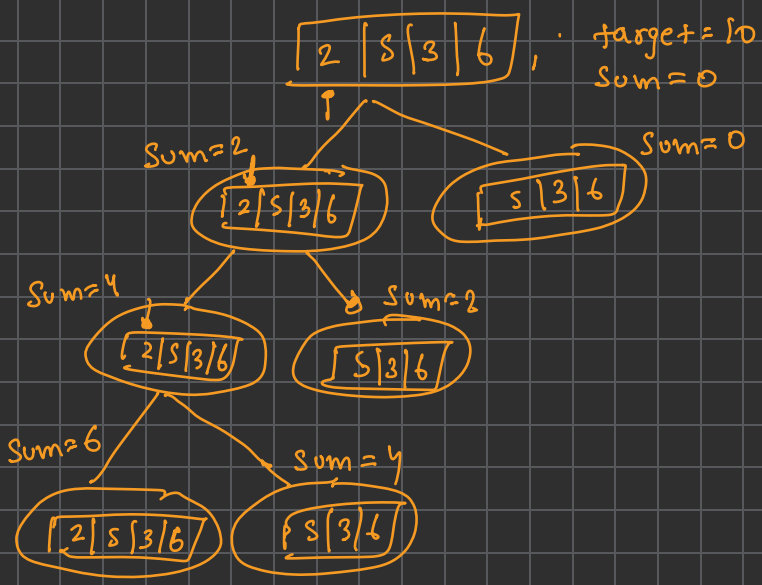
index 3
sum=7

index=4
sum=7

index=5
sum=10

```
if (sum==target)
    count++;

if (index==n || sum > target)
    return;
```




```
if (sum == target)
    count++;
    return;
if ( sum > target || index == n)
    return;
```